

```
In [1]: #pip install pandas seaborn matplotlib gradio ollama
```

```
In [2]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load Titanic Dataset
url = r"C:\Users\mahes\OneDrive - United Nations\Data Science course Naresh\3 Data
df = pd.read_csv(url)
df
```

Out[2]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.28
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.92
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.10
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.05
...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.73

891 rows × 12 columns



```
In [3]: # Display dataset info
print(df.describe())
```

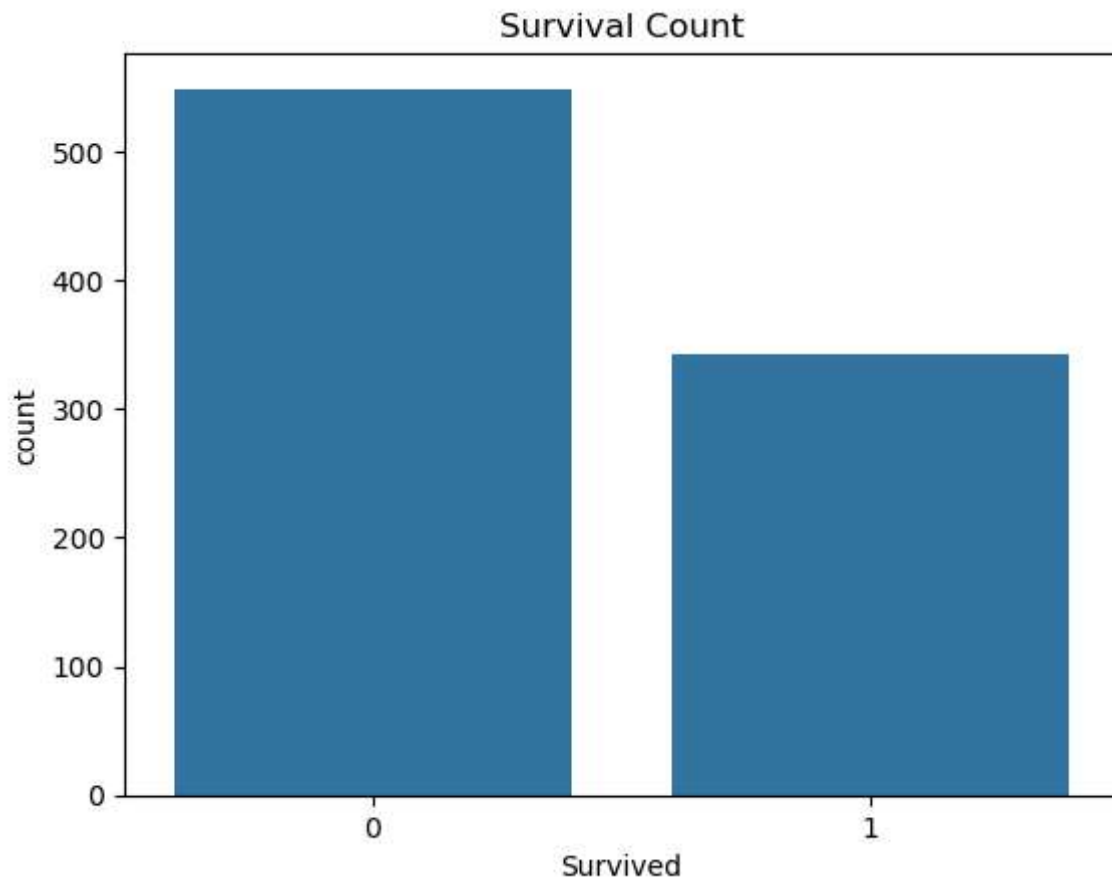
	PassengerId	Survived	Pclass	Age	SibSp \
count	891.000000	891.000000	891.000000	714.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008
std	257.353842	0.486592	0.836071	14.526497	1.102743
min	1.000000	0.000000	1.000000	0.420000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

```
In [4]: # Missing Values Check
print("\nMissing Values:\n", df.isnull().sum())
```

```
Missing Values:
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

```
In [5]: # Survival Rate Visualization
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```



```
In [6]: import ollama

def generate_insights(df_summary):
    prompt = f"Analyze the dataset summary and provide insights:\n\n{df_summary}"
    response = ollama.chat(model="mistral", messages=[{"role": "user", "content": p
    return response['message']['content']

# Generate AI Insights
summary = df.describe().to_string()
insights = generate_insights(summary)
print("\n ♦ AI-Generated Insights:\n", insights)
```

◆ AI-Generated Insights:

This dataset appears to be the Titanic passenger data, which includes information about passengers on the Titanic such as their Passenger ID, whether they survived or not (Survived), their travel class (Pclass), age, number of siblings and parents aboard (SibSp and Parch respectively), fare, and other details.

Here are some insights from the summary:

1. Total number of passengers is 891, with no missing values in any of the features.
2. The mean Survived value is approximately 0.38, suggesting that slightly more than half of the passengers did not survive the disaster.
3. The average age of the passengers was 29.69 years, but there was a significant spread with a standard deviation of 14.53 years.
4. Most passengers were traveling in 3rd class (Pclass mean of 2.31), followed by 2nd class and then 1st class.
5. The average number of siblings or spouses aboard was 0.52, and the average number of parents aboard was 0.38. This means that many passengers were traveling alone or with just one family member.
6. The average fare was \$32.20, but it ranged from \$0 to \$512.33, indicating a wide variation in ticket prices.
7. Approximately 25% of the passengers had an age of less than 20 years, 50% were between 20 and 38 years old, and 75% were below 40 years of age.
8. Only one passenger survived out of 891, suggesting that the Titanic disaster was a tragic event with very few survivors.

```
In [7]: import gradio as gr

def eda_analysis(file):
    df = pd.read_csv(file.name)
    summary = df.describe().to_string()
    insights = generate_insights(summary)
    return insights

# Create Web Interface
demo = gr.Interface(fn=eda_analysis, inputs="file", outputs="text", title="AI-Power

# Launch App
demo.launch(share=True) # Use share=True for Google Colab
```

```
* Running on local URL:  http://127.0.0.1:7860
* Running on public URL: https://8f8c3015ecb33070be.gradio.live
```

This share link expires in 1 week. For free permanent hosting and GPU upgrades, run `gradio deploy` from the terminal in the working directory to deploy to Hugging Face Spaces (<https://huggingface.co/spaces>)

AI-Powered EDA with Mistral

file



Drop File Here
- or -
Click to Upload

Clear

Submit

output

Out[7]: